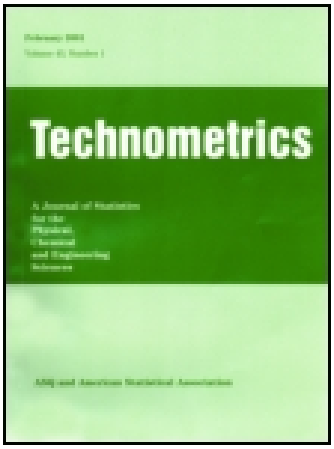
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Applied Regression Analysis and Other Multivariable Methods

Esteban Walker [a]
[a] University of Tennessee
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REFERENCES

Daniel, C. (1959), "Use of Half-Normal Plots in Interpreting Factorial Two-Level Experiments," *Technometrics*, 1, 311–341.

Nelson, W. (1969), "Hazard Plotting for Incomplete Failure Data," *Journal of Quality Technology*, 1, 27–52.

Applied Regression Analysis and Other Multivariable Methods (2nd ed.), by D. Kleinbaum, L. Kupper, and K. Muller, Boston: PWS-Kent, 1988, xviii + 718 pp.

This book updates and improves the first edition of this popular textbook (Kleinbaum and Kupper 1978). Two completely new chapters, some expanded material, and a reorganization of the chapters are the main changes in the new edition. The level and the scope were kept intact. New exercises were added to an already long list of interesting problems. Its 718 pages contain enough material for a two-semester course in applied linear models for advanced undergraduate statistics majors or nonmajor graduate students. Even though the level of ideas discussed in this book is comparable to that of Neter, Wasserman, and Kutner (1985), its avoidance of matrix algebra places it at a somewhat lower level. The book is primarily intended for people in the health, biological, and behavioral sciences; indeed, all of the examples and problems are drawn from these areas. The techniques discussed are of general application, however, so it could be used for students outside these areas.

The book's 24 chapters can be grouped into the following six topics:

1. Introduction (Chaps. 1–3): Review of basic ideas, such as levels of measurement and statistical inference.
2. Classical Regression (Chaps. 4–16): Detailed coverage of least squares estimation and normal-theory testing, ranging from simple linear regression to variable-selection techniques.
3. Analysis of Variance (Chaps. 17–20): One- and two-way designs, including blocking factors and unequal sample sizes.
4. Maximum Likelihood Methods (Chap. 21): Basic principles of maximum likelihood (ML) estimation and likelihood ratio testing with applications to Poisson and logistic regression.
5. Categorical Data Analysis (Chap. 22): The Grizzle–Starmer–Koch (GSK) approach to the analysis of multivariate nominal data.
6. Multivariate Analysis (Chaps. 23 and 24): Classical two-group discriminant analysis and exploratory factor analysis.

The introduction (Chaps. 1–3) is brief, reflecting the fact that students should have completed an introductory course. The level of measurement is nicely tied to the choice of analysis.

Chapters 4–16 present a nonmatrix discussion of least squares regression analysis. Several relevant topics are discussed in unusual detail. For example, whole chapters are devoted entirely to testing in multiple regression and to confounding and interaction. Unfortunately, the problems of estimation and prediction in the multiple regression case are ignored. Chapter 9, "Testing Hypotheses in Multiple Regression," is somewhat confusing. Here, the book refers to the standard sequential and partial F tests as variables-added-in-order and variables-added-last tests, respectively. The connection between partial F tests and t tests is not clearly made. Interestingly, the analysis of covariance (Chap. 14) is introduced from the regression point of view before the one-way analysis of variance is discussed in Chapter 17.

A new chapter (12) covers regression diagnostics. The discussion of outliers, leverage, and influence is included in the same section, which could confuse the reader. Moreover, no plots are used to illustrate these issues. The last part of this chapter examines the problem of collinearity. Unlike many other (even more advanced) books, the discussion includes the important role of the intercept (or centering) in measuring collinearity. For example, referring to the variance proportions, it is noted: "The fact that the intercept loads highly on a problem component (i.e. one with an eigenvalue near 0) would suggest to an experienced analyst that centering may help to reduce the problem" (p. 215). The chapter falls short, however, in its discussion of remedial measures (ridge and principal component regression are mentioned only in passing).

The analysis of variance material is covered in Chapters 17–20. The linear-model approach is used (without the use of matrices), with emphasis given to how different codings of the dummy variables can lead to the same tests. A drawback is the absence of graphs to illustrate, for example, the concept of interaction. Surprisingly, the use of covariates is not discussed within the context of analysis of variance.

It is a little disappointing to see this book making the mistake, made in many other textbooks, of not distinguishing between the way in which the treatments are allocated to the experimental units (experimental design) and the way in which the treatments are obtained (treatment structure). An example of a randomized complete block design with a simple factorial structure of the treatments would have helped to clarify this difference.

Maximum likelihood methods are presented in a new chapter (21). These ideas are introduced nicely without the use of heavy mathematical machinery. After working the normal case, the Poisson case is discussed, and the concept of deviance is introduced. Two detailed examples of Poisson regression are presented. In contrast, no examples or problems are given for logistic regression. Annoyingly, readers are referred to Kleinbaum, Kupper, and Morgenstern (1982) for more details.

Chapter 22 gives an ample discussion of the GSK approach to the analysis of multivariate categorical data, which uses weighted least squares estimation. What is developed here is well done and illustrated with good examples. But in several respects this material is out of date. The ML method that now dominates the analysis of categorical data is only mentioned in a small note. GENCAT, the only software advocated in this chapter, has now been incorporated into the SAS system's PROC CATMOD (which has ML capabilities as well). Finally, the reference list contains no paper published after 1977.

Chapters 23 and 24 discuss two multivariate techniques: two-group discriminant analysis and exploratory factor analysis. Even though the discussion is interesting and well illustrated, it is arguable whether these chapters fit in a book like this.

The authors' rich consulting experience is reflected in the amount of practical advice given to the applied researcher and the student. Many relevant and real examples are used to illustrate every technique. Each example is worked out in (sometimes too much) detail. A definite plus for instructors is the wealth of interesting problems at the end of each chapter. The solutions to many of these problems are given at the end of the book, and a solutions manual to all of the problems of the first edition is available. A diskette containing most of the data sets in the book is also available from the publisher. The references are, for the most part, more than adequate for a book of this level.

The book is written in a clear, understandable style, and the overall presentation of the material is attractive. Despite some drawbacks (that can be mended by instructors), this is a book that I would use to teach an applied linear models course for nonmajor graduate students, especially in the biological and health sciences. The strongest point of this book is its treatment of advanced ideas in linear models without the use of matrices. But this same lack of matrix usage makes it less suitable for advanced undergraduates in statistics.

Esteban Walker
University of Tennessee

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REFERENCES

Kleinbaum, D. G., and Kupper, L. L. (1978), *Applied Regression Analysis and Other Multivariable Methods*, North Scituate, MA: Duxbury Press.

Kleinbaum, D. G., Kupper, L. L., and Morgenstern, H. (1982), *Epidemiologic Research: Principles and Quantitative Methods*, Belmont, CA: Lifetime Learning Publications.

Neter, J., Wasserman, W., and Kutner, M. H. (1985), *Applied Linear Statistical Models* (2nd ed.), Homewood, IL: Richard D. Irwin.

Practical Statistics for Engineers and Scientists, by Nicholas P. Cheremisinoff, Lancaster, PA: Technomic Publishing, 1987, xi + 211 pp., $29.

When I received a copy of this book, I was really excited. There is a great need for books that relate the practical use of statistics to the applied areas of engineering and science. Here was such a book. Unfortunately, I was on the brink of disappointment.

Cheremisinoff begins the book appropriately enough with a chapter on "Notations and Definitions." He soon stumbles, however, during his explanation of histograms. I do not feel this explanation would be clear to someone who was not already familiar with histograms.

There are other unclear sections as well. For instance, the author jumps from the use of z tables to the use of t tables without any explanation.

I also found it very unusual that all of the tables were included in the body of the text instead of in an appendix in the back of the book. I feel that this will make it difficult to refer to these tables later. To make this placement of the tables even more distracting, they sometimes were located several pages away from the section where their use was discussed.

These faults would not be fatal, but Cheremisinoff also makes some statements at which I as a statistician balk. For example: "Select the risk alpha of asserting that a difference exists when there is no difference; i.e., Ho: $\mu > \mu_0$, Ho: $\mu < \mu_0$" (p. 28). Is this supposed to reflect a null hypothesis versus an alternate hypothesis? Why then are they both designated as "Ho"? What does a one-sided hypothesis test have to do with the statement immediately preceding it? In fact, the author consistently uses the notation "Ho" throughout the book to designate both the null hypothesis and the alternate hypothesis.

In an example in which the calculated value of the test statistic does not allow him to reject the null hypothesis, Cheremisinoff states that "more data is needed to make an assessment" (p. 82). In another example, after calculating the sample size required to estimate a parameter within 10% and arriving at a required sample size of 189, he states, "Obviously we have selected too stringent a requirement" (p. 24). I find both of these statements dangerous. They seem to suggest that if we do not like the results we can question the method used to achieve them. The time to question the method is before the data are collected.

The sections on spectral analysis are different from all other sections. Here Cheremisinoff goes into much more depth, uses a lot of formulas, and delves deeply into theory. In light of the intent stated in the preface to "not provide rigorous treatment of statistical theorems and theory" but rather "a handy reference of standard statistical analysis and data regression techniques" (p. ix), I find this theoretical treatment hits a false note.

Although my list of concerns continues, I will stop here. In all fairness, there were some good aspects of the book, such as a chapter on control charts, but I feel the deficiencies considerably outweigh the good aspects. Novices deserve to have things explained clearly and correctly. This book errs on both counts. I would not recommend it to anyone.

Annabeth L. Propst
Process Management Institute

Design, Data, and Analysis, edited by Colin L. Mallows, New York: John Wiley, 1987, xix + 380 pp., $29.95.

This book, subtitled "by Some Friends of Cuthbert Daniel," was written as an 80th birthday testimonial to the influence of Daniel's work on a lot of people who designed experiments and analyzed data. It has been lovingly assembled by the editor and primarily contains papers that describe experiments, give all of the data, and show the analysis. There also are a few expository papers. They are all worthy of a listing with some description of their content.

1. "Analysis of Unreplicated Factorials Allowing for Possibly Faulty Observations," by G. E. P. Box and R. Daniel Meyer—This paper illustrates the type of pattern that can occur on a probability plot when a factorial experiment has a bad value. Posterior probabilities are then used to quantify this type of effect.

2. "Variable Selection in Clustering and Other Contexts," by Edward B. Fowlkes, Ramanthan Gnanadesikan, and Jon B. Kettenring—Variable selection in multiple regression and in discriminant analysis are the other contexts. A discussion of algorithms is followed by an example that uses all three procedures to elucidate various relationships among the variables.

3. "Weighing Qualitative Data," by Spencer M. Free—This very short paper discusses some data collected to determine a vaccination procedure for some cattle to protect them from a virus.

4. "Explaining Experimental Design Fundamentals to Engineers: A Modern Approach," by Gerald J. Hahn, John L. Bemesderfer, and Donald M. Olsson—The first part of this very long paper explains the objectives and philosophy that were combined to create a training course at General Electric. There follows an extensive discussion of the material that was presented in each of seven sessions. Some of the examples that were used are included.

5. "Repeatability and Reproducibility: Some Problems in Applied Statistics," by Hugo C. Hamaker—The author contends that the data from the usual interlaboratory study for a test method should be analyzed by regression, not by analysis of variance. He explains that the sample by laboratory interaction has no part in the analysis and that transformed data are useless in this situation. Two examples illustrate the first contention, and the results are used for the regression analysis that replaces transformations.

6. "Design, Modelling, and Analysis of Some Biological Data Sets," by Frank R. Hampel—Eight different data sets are considered. The author states that the data were part of his consulting and that his analysis was only what was required by the data and their context. The paper is discursive, and the discussions for each example are extensive.

7. "Analytical Representation of Dielectric Constants: A Complex Multiresponse Problem," by Stephen Havriliak and Donald Watts—Multiresponse parameter estimation is used to estimate the parameters of structure–property relationships of polymers. The results are compared to graphical estimation methods. Extensive use is made of residuals analysis.

8. "Indirect Measurement of Work," by Mary B. Hovik and Sutton Monro—The concern is the office or shop, and the data come from accounting records. The objective is to estimate the man-hours per unit produced for different work categories. The paper is very short.

9. "Are Some Latin Squares Better Than Others?" by J.

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